THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–14) | Class 8 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** 49	$7 \times 7 = 49$
2	**(a)** 2^7	Add the exponents: $3 + 4 = 7$
3	**(a)** 17	Add the values of the symbols, subtracting a smaller symbol placed before a larger one
4	**(c)** 90°	The angles of a quadrilateral add up to 360°
5	**(b)** 2346	digit sum 15 is a multiple of 3
6	**(d)** 1287	$99 \times 13 = (100 - 1) \times 13$
7	**(b)** 2 : 3	Divide both terms by 6
8	**(c)** 25%	Multiply by 100: $\frac{1}{4} \times 100 = 25$
9	**(b)** 5	$3^2 + 4^2 = 9 + 16 = 25$, hypotenuse $= 5$
10	**(c)** 2:3	Divide both terms by HCF 6
11	**(c)** 3	Each stage triples the count: $3^1 = 3$
12	**(c)** 21	Sum $= 105$; $\div 5$
13	**(a)** $2x + 5$	2 times x is $2x$; add 5
14	**(b)** 84 cm²	12×7
15	**(d)** 144	$12 \times 12 = 144$
16	**(d)** 2^5	Subtract the exponents: $9 - 4 = 5$
17	**(a)** VIII	8 is written as VIII
18	**(d)** 115°	Adjacent angles of a parallelogram add up to 180°
19	**(a)** 4527	digit sum 18 is a multiple of 9
20	**(a)** 91	$7 \times 10 + 7 \times 3$

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	$27^2 = 27 \times 27 = 729$ (2).	2
22	Add exponents: 2^5 (1); $= 32$ (1).	1+1
23	27 = XXVII (2).	2
24	Sum of angles $= 360^\circ$ (1); fourth angle $= 360 - 270 = 90^\circ$ (1).	1+1
25	Yes (1); digit sum 15 is a multiple of 3 (1).	1+1
26	$99 \times 16 = (100 - 1) \times 16$ (1) $= 1600 - 16 = 1584$ (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	$196 = 2^2 \times 7^2$ (1); pairing equal factors (1); $\sqrt{196} = 14$ (1).	1+1+1
28	$2^5 \times 2^4 = 2^9$ (1); $2^9 \div 2^6 = 2^3$ (1); value 8 (1).	1+1+1
29	XLVII = 47 (1); CLXIV = 164 (1); add the symbol values, subtract when a smaller symbol comes before a larger one (1).	1+1+1
30	$\angle B = 180 - 62 = 118^\circ$ (1); $\angle C = \angle A = 62^\circ$ (1); $\angle D = \angle B = 118^\circ$ (1).	1+1+1
31	By 2: Yes, last digit 0 is even (1); by 5: Yes, last digit 0 is 0 or 5 (1); by 10: Yes, last digit 0 is 0 (1).	1+1+1
	OR: $2x + 6 + 4x + 20$ (1); collect like terms $= 6x + 26$ (2).	1+2
32	Total parts 8 (1); one part ₹ 150 (1); shares ₹ 450 and ₹ 750 (1).	1+1+1
	OR: Profit = ₹ 50 (1); percent = 50/200×100 (1) = 25% (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	$180 = 2^2 \times 3^2 \times 5$ (1); primes with odd power: multiplier $= 5$ (2); product $= 900$ (1); square root $= 30$ (1).	1+2+1+1
34	(i) 32 (1); (ii) $\frac{1}{9}$ (1); (iii) 1 (1); (iv) sum $= 32 + \frac{1}{9} + 1 = \frac{298}{9}$ (2).	1+1+1+2
35	12 = XII (1); 25 = XXV (1); 38 = XXXVIII (1); 46 = XLVI (1); 59 = LIX (1).	1+1+1+1+1
36	Fourth angle $= 95^\circ$ (2); exterior angles $180 - 70 = 110^\circ$, 70°, 95°, 85° (2); their sum $= 360^\circ$ (1).	2+2+1
	OR: By 2: Yes, last digit 0 is even (1); By 3: Yes, digit sum 18 is a multiple of 3 (1); By 5: Yes, last digit 0 is 0 or 5 (1); By 9: Yes, digit sum 18 is a multiple of 9 (1); By 10: Yes, last digit 0 is 0 (1);	1+1+1+1+1
37	$x^2 + 4x + 2x + 8$ (2); $= x^2 + 6x + 8$ (1); at $x = 3$: $9 + 18 + 8 = 35$ (2).	2+1+2
	OR: Total parts 8, one part $= 6$ (2); numbers 18 and 30 (2); difference 12 (1).	2+2+1
38	Food = 30% of 20000 = ₹ 6000 (2); rent = 25% of 20000 = ₹ 5000 (2); left = 20000 − 11000 = ₹ 9000 (1).	2+2+1
	OR: Diagonal$^2 = 9 + 16 = 25$, diagonal $= 5$ m (3); distance along two sides $= 7$ m, saved $= 2$ m (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–14) | Class 8 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** 12	Multiply both terms by 4
2	**(d)** $\frac{1}{5}$	$20\% = \frac{20}{100} = \frac{1}{5}$
3	**(d)** 4	$5^2 - 3^2 = 16$, other leg $= 4$
4	**(c)** 12	Multiplier 4, so $x = 3 \times 4$
5	**(b)** 5	$n + 2 = 5$ (two bases and n rectangles)
6	**(d)** 28	Arrange in order; take the middle value
7	**(c)** 19	$3 \times 5 + 4$
8	**(a)** 30 cm²	$\frac{1}{2} \times 10 \times 6$
9	**(c)** 13	$13 \times 13 = 169$
10	**(d)** 2^6	Multiply the exponents
11	**(b)** 5	Add the powers of 2 where there is a 1
12	**(c)** 58°	Opposite angles of a parallelogram are equal
13	**(d)** 3846	last digit 6 is even
14	**(c)** $3x + 12$	Multiply each term inside by the outside number
15	**(c)** 2 : 3 and 8 : 12	Product of extremes $= 2 \times 12$ equals product of means $= 3 \times 8$
16	**(d)** 25	10/100 × 250
17	**(c)** 3, 4, 5	$3^2 + 4^2 = 25 = 5^2$
18	**(a)** 250	One part $= 400 \div 8 = 50$
19	**(a)** 4	n base vertices + 1 apex $= 4$
20	**(a)** 18	Most frequent value

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	2 m = 200 cm (1); ratio $= 1 : 5$ (1).	1+1
22	$\frac{3}{8} \times 100 = 37.5\%$ (2).	2
23	$h^2 = 3^2 + 4^2 = 9 + 16 = 25$ (1); $h = 5$ cm (1).	1+1
24	Cost of 1 pen $=$ ₹15 (1); cost of 15 pens $=$ ₹225 (1).	1+1
25	Faces $= n + 2 = 5$ (1); vertices $= 2n = 6$, edges $= 3n = 9$ (1).	1+1
26	Sum $= 140$ (1); mean $= 140 \div 5 = 28$ (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Height$^2 = 5^2 - 3^2 = 25 - 9 = 16$ (2); height $= 4$ m (1).	2+1
28	Actual distance $= 5 \times 12 = 60$ km (2); map distance $= 96 \div 12 = 8$ cm (1).	2+1
29	Faces $= 4$, vertices $= 4$, edges $= 6$ (2); $F + V - E = 4 + 4 - 6 = 2$ (1).	2+1
30	Mean $= 270 \div 6 = 45$ (2); students above mean $= 2$ (1).	2+1
31	$3 \times 6 + 4 \times 2 - 5$ (2) $= 21$ (1).	2+1
	OR: Area $= bh = 126$ cm² (2); triangle $= \frac{1}{2} \times 126 = 63$ cm² (1).	2+1
32	$512 = 2^9$ (1); grouping equal factors in triplets (1); $\sqrt[3]{512} = 8$ (1).	1+1+1
	OR: $4.2 \times 10^5 = 420000$ (1); $3.5 \times 10^{-3} = 0.0035$ (1); move the decimal point right for a positive power and left for a negative power (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Angle per student $= 360^\circ \div 30 = 12^\circ$ (1); Cricket: 144°, Football: 108° (2); Tennis: 72°, Hockey: 36° (1); sum $= 360^\circ$ (1).	1+2+1+1
34	Stage 1: 4 segments of 3 cm (1); stage 2: $4 \times 4 = 16$ segments (1); each of length $= 1$ cm (1); total $= 16 \times 1 = 16$ cm (2).	1+1+1+2
35	Mean $= 15.17$ (1); median $= 15.5$ (2); mode $= 25$ (1); range $= 21$ (1).	1+2+1+1
36	$3n + 5 - 5 = 3n$ (3); for $n = 4$, result $= 12$ (2).	3+2
	OR: Side $= 24 \div 4 = 6$ m (1); area $= 6^2 = 36$ m² (2); tiles $= 36$ (2).	1+2+2
37	$108 = 2^2 \times 3^3$ (1); smallest divisor $= 4$ (2); quotient $= 27$ (1); cube root $= 3$ (1).	1+2+1+1
	OR: $(2^2)^3 = 2^6$ (1); divide: 2^2 (1); multiply: 2^3 (2); value 8 (1).	1+1+2+1
38	Steps: $19 = 2 \times 9 + 1$; $9 = 2 \times 4 + 1$; $4 = 2 \times 2 + 0$; $2 = 2 \times 1 + 0$; $1 = 2 \times 0 + 1$ (3); reading remainders from last to first gives 10011_2 (2).	3+2
	OR: $\angle B = 114^\circ$ (1); $\angle C = \angle A = 66^\circ$ (1); $\angle D = \angle B = 114^\circ$ (1); perimeter $= 4 \times 7 = 28$ cm (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–14) | Class 8 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** $6x + 8$	Add like terms
2	**(b)** 36 cm²	base × height $= 9 \times 4$
3	**(c)** 64	$4 \times 4 \times 4 = 64$
4	**(b)** $\frac{1}{16}$	$a^{-n} = \frac{1}{a^n} = \frac{1}{16}$
5	**(d)** 101	Divide repeatedly by 2 and read the remainders upwards
6	**(a)** 24 cm	$4 \times 6 = 24$
7	**(c)** 21	Add all the digits
8	**(d)** $a^2 + 2ab + b^2$	Standard identity
9	**(c)** 1 : 4	Convert to the same unit first
10	**(d)** 550	Increase = 50; new = 550
11	**(c)** 12	Hypotenuse $= 5$; perimeter $= 3 + 4 + 5 = 12$
12	**(b)** 35	5×7
13	**(b)** 9	$3n = 9$
14	**(b)** 33	Highest − lowest
15	**(c)** $10x + 2$	Place value: tens × 10 + ones
16	**(d)** 35 cm²	$\frac{1}{2}(8 + 6) \times 5$
17	**(d)** 4	$4^3 = 64$
18	**(a)** 3.2×10^3	One non-zero digit before the decimal point
19	**(d)** 600	Place value = digit x value of its position
20	**(b)** 26 cm	$2(8 + 5)$

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	$5n + 7$ (2).	2
22	Area $= l \times b = 25 \times 12$ (1) $= 300$ m² (1).	1+1
23	Yes, $225 = 15 \times 15$ (1+1).	1+1
24	$2^{-6} = \frac{1}{2^6}$ (1) $= \frac{1}{64}$ (1).	1+1
25	$1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 = 7$ (1+1).	1+1
26	Opposite angle 48° (1); adjacent angles $180 - 48 = 132^\circ$ each (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	$4 \times 10^3 + 0 \times 10^2 + 8 \times 10^1 + 6 \times 10^0$ (3).	3
28	BD = AC = 16 cm (1); OA = half of AC = 8 cm (1); OB = 8 cm (1).	1+1+1
29	Digit sum of the other digits $= 7$ (1); need a total that is a multiple of 3 (1); digits: 2, 5, 8 (1).	1+1+1
30	$x^2 + 3x + 2x + 6$ (2); $= x^2 + 5x + 6$ (1).	2+1
31	Product of extremes 3x20$= 60$ (1); product of means 4x15$= 60$ (1); equal, so in proportion (1).	1+1+1
	OR: Discount = 10/100×1200 = ₹ 120 (2); SP = 1200 − 120 = ₹ 1080 (1).	2+1
32	Hypotenuse$^2 = 25 + 144 = 169$, hypotenuse $= 13$ cm (2); perimeter $= 5 + 12 + 13 = 30$ cm (1).	2+1
	OR: Total parts $= 10$, one part $=$ ₹120 (1); A = ₹ 240 (1); B = ₹ 360, C = ₹ 600 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 6 × 5 = 30

Q.	Expected Answer / Value Points	Marks

33	Remainder when 4537 is divided by 9 is 1 (2); number to add $= 9 - 1 = 8$ (2); new number 4545 (1).	2+2+1
34	$104 \times 25 = (100 + 4) \times 25 = 2500 + 100$ (2); $= 2600$ (1); direct multiplication gives 2600 (2).	2+1+2
35	Cost of 1 pen ₹ 15 (1); cost of 10 pens ₹ 150 (2); pens for ₹ 210: 14 (2).	1+2+2
36	(i) SP = 2000×110/100 = ₹ 2200 (2); (ii) SP = 2000×95/100 = ₹ 1900 (2); difference = ₹ 300 (1).	2+2+1
	OR: Triplet: $4, 3, 5$ (2); $16 + 9 = 25$ (2); $5^2 = 25$ so verified (1).	2+2+1
37	Total food $= 20 \times 15 = 300$ person-days (2); people now $= 30$ (1); days $= 300 \div 30 = 10$ (2).	2+1+2
	OR: Stage 1: 3 (1); stage 2: 9 (1); stage 3: 27 (1); each stage multiplies the number by 3 (2).	1+1+1+2
38	Sum of five $= 5 \times 20 = 100$ (2); sum of four $= 85$ (1); fifth $= 15$ (2).	2+1+2
	OR: Middle row: 2+5=7 (1), 5+3=8 (1); top $= 7 + 8 = 15$ (3).	1+1+3

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